



UNIVERSITY OF
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Superconductivity

Lecture 02 – “Perfect Expulsion”

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'Surprising' observation

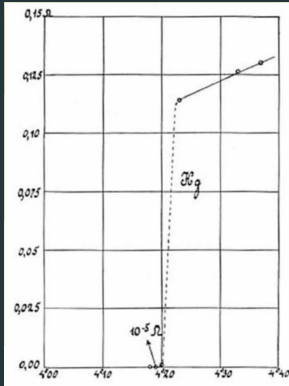


Figure 1: Electrical



Figure 2: Magnetic

- Superconductivity is an 'electro-magnetic' phenomenon.
- Shouldn't we be able to 'describe' using Maxwell's equations?

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Magnetism of a 'perfect' conductor

How to measure R , where $R \rightarrow 0$

The question is essentially concerned with increasing precision in the resistance R measurement: $R < 0.1$, $R < 0.01$, $R < 0.001$...

- Flow a current, measure a voltage drop ($V = IR$)
- Can't indefinitely increase I (due to a limit by critical current I_c)
- Can't indefinitely increase the length of a wire

Prepare a ring

Instead, one could measure the effects arising from R , for instance, dissipation.

- $RI + L \frac{dI}{dt} = 0$
- $I(t) = I(0) \exp[-(R/L)t]$
- How to setup the current initially?
- How to measure $I(t)$ without interruption?

A 'perfectly conducting' ring

From the Lenz law,

$$RI + L \frac{dI}{dt} = - \frac{d\Phi}{dt}, \quad (1)$$

where the magnetic flux $\Phi = S \cdot B$ (S being an enclosed area).

For $R = 0$

$$\frac{d}{dt} (SB + LI) = 0. \quad (2)$$

- What does each term inside the parenthesis above represent?
 - 'applied flux' (SB) + 'induced flux' (LI)
- What does it imply?
 - a persistent current I is set up,
 - the total magnetic flux remains 'constant'.

How it looks like

Note that flux conservation concerns only total flux, but not its vectorial density distribution.

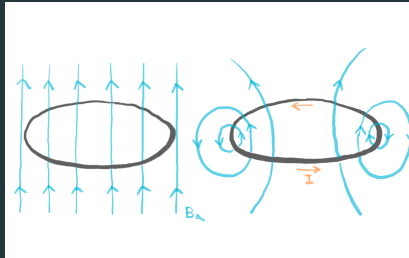


Figure 3: Left: magnetic flux density as applied at $T > T_c$ then cooled below T_c . Right: when the applied flux density was removed and only the induced flux density and the resulting currents remaining.

Literally persistent

- No field change over a few years
- Translating to at most $\rho = 10^{-26}$ Ohm m
- 18 orders of magnitude smaller than ρ (300 K) of Cu
- Zero-resistance state is as much distinct from a good metal as a good metal is distinct from insulator.

A perfectly conducting bulk

Draw in mind many arbitrary rings inside a bulk, then one could expect

- $\partial \mathbf{B} / \partial t = 0$
- $\mathbf{E} = 0$ (from Faraday's induction law)
- consistent with $E = 0$ for $\rho = 0$ (Ohm's law)

Consequence of having a perfectly conducting bulk

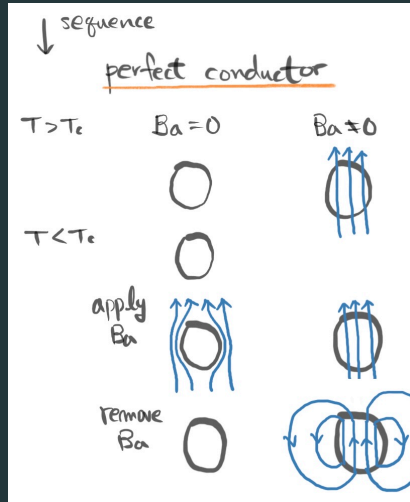


Figure 4: A magnetic field response of a perfect conductor as expected.

Magnetism of a 'super'conductor

It is observed, however, for a 'super-'conductor

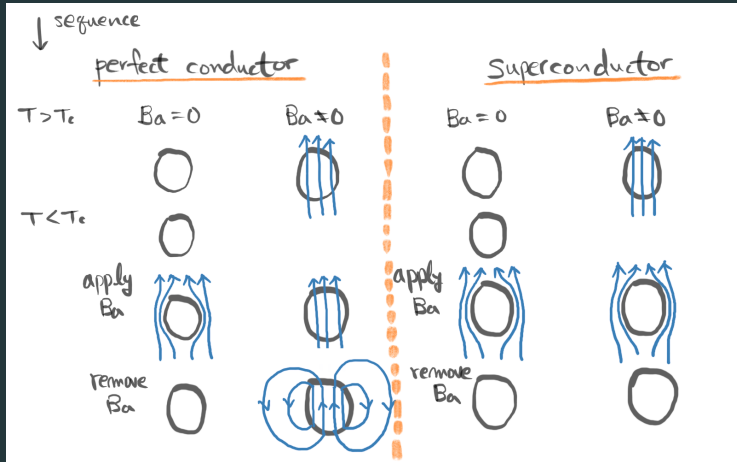


Figure 5: A magnetic field response of a perfect conductor as expected (left) and a superconductor as observed(right).

For a superconductor 1/2

- Not just $B = \text{const.}$ as $\partial \mathbf{B} / \partial t = 0$, but
- $B = \text{const.} = 0$.

What does this imply?

Recall

$$\mathbf{B} = \mu_0 (\mathbf{H}_a + \mathbf{M}), \quad (3)$$

where $\mathbf{H}_a$ is an applied magnetic field intensity and $\mathbf{M}$ the magnetisation induced in material.

Since $B = 0$ for a superconductor, one obtains

$$\mathbf{M} = -\mathbf{H}_a. \quad (4)$$

Recall a definition $\mathbf{M} = \chi \mathbf{H}_a$, where χ is called the magnetic susceptibility.

For a superconductor 2/2

$$\mathbf{M} = \chi \mathbf{H}_a. \quad (5)$$

with $\chi = -1$ implies two things:

- the minus sign indicates 'dia-magnetism' and
- the value of unity indicates 'perfectness'

So 'perfect diamagnetism' (also perfect expulsion of magnetic flux).

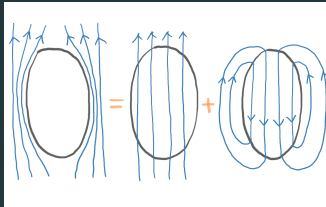


Figure 6: Net flux as a sum of applied field and flux from magnetisation.

Penetration

Perfect up to a small penetration depth

- Sample magnetisation in bulk = surface currents.
- The current cannot be confined to an infinitesimal thickness (as unphysical $j \rightarrow \infty$).
- So allow some very small (e.g., a few micron) yet finite thickness where $B = 0$ ceases to apply.

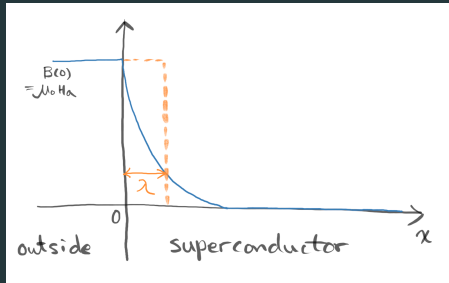


Figure 7: Exponential suppression of a magnetic flux when entering a superconducting bulk.

Perfect up to a small penetration depth

It was found that the penetration depth λ follow empirically

$$\lambda = \frac{\lambda_0}{\sqrt{1 - (T/T_c)^4}}. \quad (6)$$

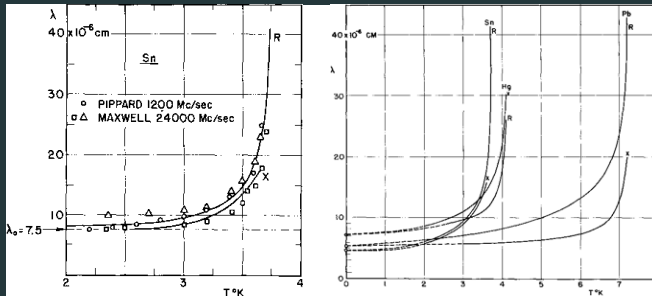


Figure 8: Penetration depth as a function of temperature for several elemental metals [Schawlow and Devlin, Phys. Rev. 113, 120 (1959)].